Lean evidence inventory — 2026-10-21

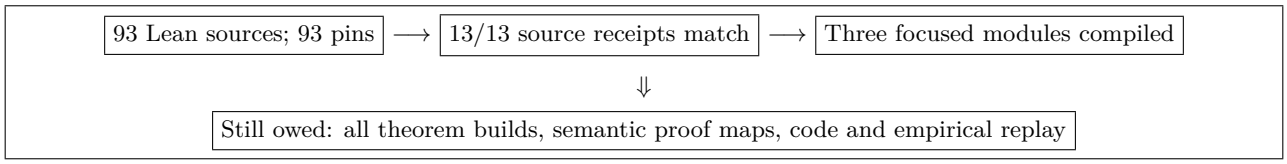
cryptanalysis main `d1a10f6d537dc4e9b5aff02c1e4dcef3e2267c1b`

Scope. This change audits tracked Lean source metadata, adds a root toolchain pin, and checks source-bound proof receipts. It does not build every theorem or semantically review every experiment across the three requested repositories. No scientific trial or performance measurement was run.

Inventory and correction

There are 93 tracked Lean files. Before this change, 33 inherited no `lean-toolchain`. The new root pin is `leanprover/lean4:v4.34.1`, while local pins still win. After the change, 92 files resolve to Lean 4.34.1 and the binary-genus-2 module retains Lean 4.24.0.

The new `tools/audit_lean_evidence.py` reads committed files even in sparse checkouts. It checks each pin, scans Lean code outside comments and strings for forbidden proof tokens, and compares dedicated Lean receipt source SHA-256 values and toolchain claims. The observed result was **0 missing pins, 0 files with forbidden tokens, and 13/13 matching source receipts**. CI now runs this static check on relevant changes. A pin, lexical scan, or matching receipt is not a kernel theorem verdict.



Two files have no local `#print axioms`: OrbitPairIndex is covered by its sibling `AxiomAudit.lean`; NearestSelector is imported through `ProofAudit.lean`. The scan reports them for separate review instead of treating file presence as proof. Five parser controls for comments, strings, split tokens, and a real `sorry` passed.

Live proof checks and limits

Check	Observed result and boundary
Lean 4.34.1 OrbitCompression	PASS; four reported declarations, using only standard foundations on the join-work formula. Abstract cost model only.
Lean 4.34.1 GaugeTree	PASS; five reported declarations, using only `propext`. Explicit group and order-three automorphism premises.
OrbitPairIndex	Built and axiom-audited in merged cryptanalysis PR 1008, with four independent Python tests. Only the modular lookup bridge is proved.
Dedicated receipts	13/13 source hashes match; original hardware and all proof modules were not replayed.
Binary-genus-2 and remaining modules	Pinned or inventoried, but not built in this change.

No corrected scientific measurement arose from this audit. The checked OrbitCompression result itself leaves complete-operation wall time for an isolated host. The existing orbit reference JSON and performance archives were not rewritten; this report's diagram is an evidence visual, not a performance graph.

Requested scope and status

Requirement	Status
Faithful Lean proofs for every historical experiment across three repos	Partial. Pins and receipts checked; three representative modules built across the prior and current slices. Exhaustive builds and semantic maps remain open.
Legitimate code and numerical results	Partial. Two existing proof models replayed here; complete implementation and empirical replay remain.
Graphs and performance dashboards	Partial. Prior autoresearcher PR refreshed nine N13 rows; this change adds an evidence diagram and no new performance number.
Everything good to go	Not established. Full kernel, code-to-model, empirical, and cross-repository reviews remain.

Reproduce the static audit with `python3 tools/audit_lean_evidence.py`; use its `-json` option to save the complete result. The separate `diagram.svg` shows the same proof boundary.